Credit Risk Scoring: Executive Business Report

German Credit data, 1,000 historical loans (Good/Bad credit-risk labels, not confirmed defaults) | Every number comes from one run of the accompanying notebook

1. Summary

A tuned **Random Forest** ranks risky applicants well on 200 held-out loans: ROC-AUC **0.81** (95% bootstrap interval 0.74 to 0.88) against 0.72 for a simple decision tree. On the training split alone, a nested cross-validation gives 0.79 +/- 0.05 across folds; this is a descriptive check of the tuning procedure, not independent confirmation. Missing a bad applicant costs five times as much as declining a good one. The cutoff chosen from nested out-of-fold training predictions (0.16) is therefore very strict: it flags 93% of bad-labelled test cases but also declines 71% of good ones (approval rate 22%). Expected cost is **0.60 per applicant versus 1.50 if everyone were approved**. The cost curve is flat, so the real decision is growth versus risk. The main drivers are the checking-account status and the size and length of the loan. We recommend a controlled pilot, not automation.

2. What the data says

30% of the loans carry the bad label (700 good, 300 bad); nothing is missing. About 7% of loan amounts and of durations are IQR outliers; they were kept, and the notebook does not test whether bad labels concentrate among them. The strongest raw signals are the **checking-account status** and the **duration**.

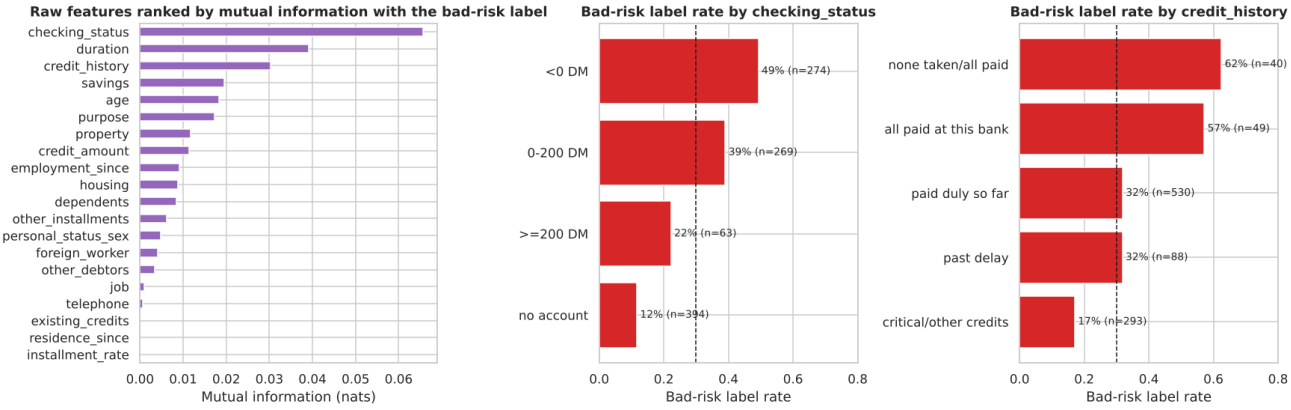


Figure 1. Left: raw features ranked by association with the bad label. Middle and right: bad-label rate by checking account and by credit history (dashed line = 30%).

The bad rate climbs from 12% (no checking account) to 49% (overdrawn). Credit history is counter-intuitive: "critical/other credits" is the safest group (17%), "none taken/all paid" the worst (62%, n=40). We read this as an effect of past lending choices, not a rule to copy.

3. Models, cutoff and calibration

Preprocessing, tuning, the cutoff and the calibration use training data only; the test set is used once for evaluation. The cutoff minimises the dataset's cost matrix (missing a bad case = 5, declining a good one = 1) on **nested** out-of-fold scores: in each outer fold the grid search sees only that fold's training part.

Model	Cutoff	ROC-AUC	Avg prec.	Caught	Precision	Missed	Refused	Cost
Decision tree (depth 4)	0.21	0.72	0.49	83%	40%	10	76	0.63
Random forest (tuned)	0.16	0.81	0.69	93%	36%	4	100	0.60
Random forest at 0.50	0.50	0.81	0.69	38%	79%	37	6	0.95

Test set: 200 applicants, 60 bad-labelled. Caught = bad cases flagged; Missed = bad cases approved; Refused = good applicants declined; Cost = (5 x Missed + Refused) / 200. Approving everybody costs 1.50.

Cutoff	Approval rate	Bad cases flagged	Good applicants refused	Cost / applicant
0.16	22%	93%	71%	0.60
0.18	24%	92%	69%	0.61
0.25	40%	87%	48%	0.54
0.30	52%	80%	35%	0.55
0.40	72%	62%	14%	0.68
0.50	86%	38%	4%	0.95

Sensitivity check on the test set (not used to choose the cutoff): cost is lowest around 0.25 to 0.30 (0.54), and the nested cutoff (0.16) is stricter than an earlier non-nested estimate (about 0.23). With 200 rows these differences are noise; re-select the cutoff on a larger validation sample and with the bank's own costs.

Probabilities. The raw forest is under-confident (calibration slope 1.75, ideal 1.0). A sigmoid (Platt) calibration fitted on the nested training scores improves it (slope 1.08; Brier 0.160 to 0.152; calibration error 0.085 to 0.044). It has its own cutoff (0.12) and almost the same test cost (0.595 vs 0.600; approval 26%). If scores are shown as "percent risk", use the calibrated version.

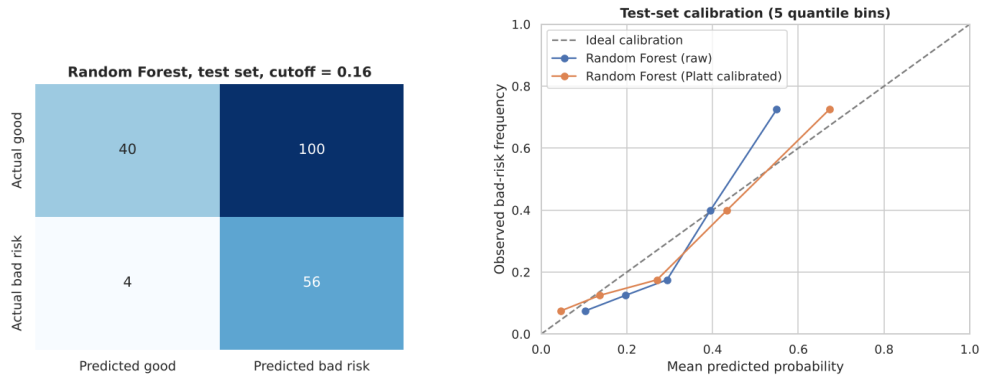


Figure 2. Left: confusion matrix at the nested cutoff (test). Right: reliability of raw and calibrated probabilities (5 bins, 200 rows).

4. Why the model decides what it decides

SHAP splits each risk estimate into the contribution of every input (values add up to the model output; computed for all 200 test records). The biggest drivers are the **checking-account status** (mean |SHAP| 0.086), the **size and length of the loan** (duration, amount and payment proxy together 0.050) and **savings** (0.026). Sex/marital status and foreign-worker status are not model inputs; dropping them left training CV AUC unchanged (0.793 vs 0.794).

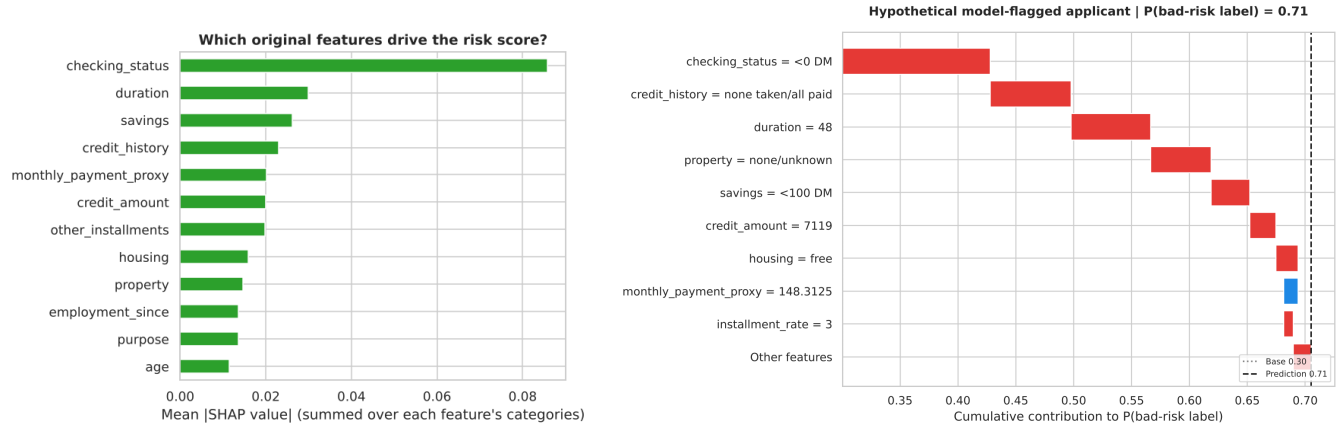


Figure 3. Left: average impact of each input. Right: one historical test record explained.

Reading one decision (plain language). The *hypothetical model-flagged applicant* is a historical test record, not a real refusal. Its estimated bad-risk probability is 71% (average 30%). Mostly because the current account is overdrawn (+0.13), the credit history, the 48-month term, unknown property and low savings. The notebook turns the reviewed factors into a letter with an LLM prompt built from this real SHAP output. Only checking account, savings and loan size/length are passed on; credit history and property are held back as possible proxies. The three next steps (avoid overdrafts, build small savings, ask for a smaller or shorter loan) are **written into the prompt in advance**, so an LLM answer mostly rephrases them. A live Gemini call, run by the author and pasted into the notebook, returned a 156-word hypothetical letter that passed the automated checks (no technical terms, three bulleted steps, hypothetical disclaimer); a person must still review it. An optional five-role LLM review council, also run by the author, found no code defect and made two low-value suggestions; it is a second opinion from one small model, **not evidence**. LIME was run by the author as a sanity check next to SHAP.

Fairness check. Group differences are small relative to the sample size.

Comparison (test set, cutoff 0.16)	Approval rate	Good applicants wrongly flagged	p (Pearson / Fisher)
Female div/sep/mar (n=61) vs male single (n=118)	20% vs 25%	30/41 (73.2%) vs 57/84 (67.9%)	0.544 / 0.679
Age 18-24 (n=39) vs 35-49 (n=63)	10% vs 27%	17/20 (85.0%) vs 34/50 (68.0%)	0.148 / 0.234

Two-sided, unadjusted p-values computed after viewing the audit table (Pearson chi-square without continuity correction; Fisher exact for small counts). No difference is statistically distinguishable, and this is neither evidence for nor against discrimination. What stands out is that at this strict cutoff 65% to 85% of good applicants are flagged in every group, and the model under-predicts risk for ages 18-24 (predicted 37.8% vs observed 48.7%). Age is still an input; a fairness and legal review is needed.

5. Cause versus correlation, and next steps

Levers the bank controls (they change repayment burden): loan amount and term, the instalment-to-income cap, credit limits and pricing, co-applicants; test them in a small randomised pilot. **Signs, not levers:** credit history (the "critical account" group is safest most likely because of past lending choices), property, housing, job tenure and age (stand in for income and wealth, which are absent), and the approved-only data (rejected applicants are missing). Do not turn these proxies into policy or customer advice.

- **Limits.** 200 test rows (AUC 0.74 to 0.88), Good/Bad labels instead of confirmed defaults, the 5:1 cost ratio from the public dataset, no dates for time-based validation, no income data.
- **Pilot first.** Shadow mode next to current decisions, then a limited live pilot monitoring calibration, drift and group error rates; every customer letter reviewed by a person.
- **Run status.** One fresh top-to-bottom run, no errors. The *shap* package was unavailable offline, so exact TreeSHAP values came from the notebook's built-in routine (additivity error 3.6e-14) and the SHAP figures are matplotlib renderings. LIME, the Gemini letter and the council ran in the author's own session; their outputs are pasted in the notebook.